

Causal Inference and ML: Modeling Coping Strategy Improvement

Estimating the influence of the socioeconomic environment on individual coping strategies

Github Repository:

github.com/marthab-oss/Modeling-Coping-Strategy-Improvement-via-Causal-Inference

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Examination of the influence of different variables on individual Coping Skill:

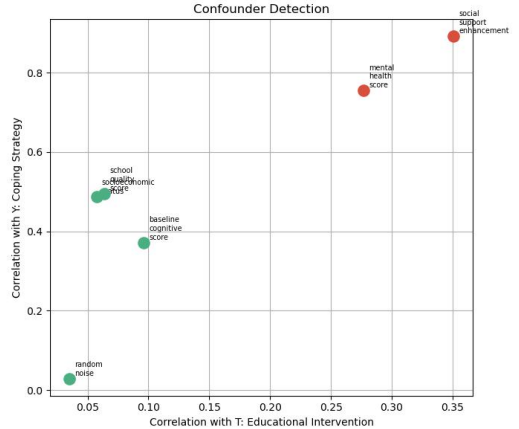
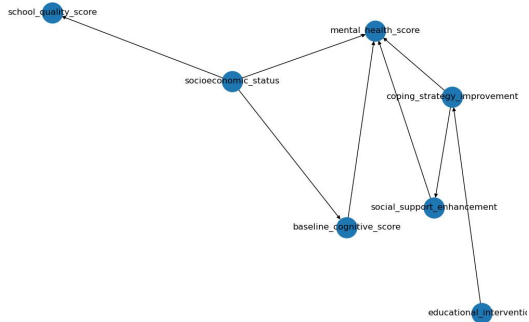
- Does the socioeconomic environment introduce selection bias in the assignment of Educational Intervention?
- Which covariables have the strongest predictive and causal influence on Coping Strategy Improvement?
- Does the Educational Intervention affect all individuals equally, or are there heterogeneous treatment effects depending on individual characteristics?

Data Structure - Dataset Excerpt

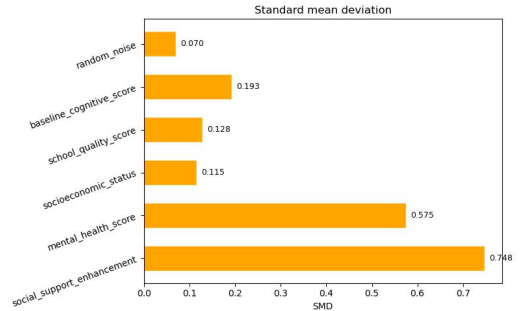
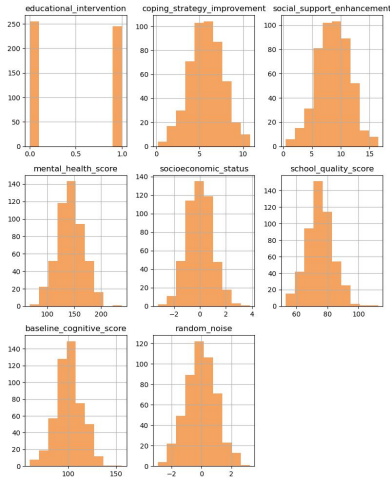
educational_intervention	coping_strategy_improvement	social_support_enhancement	mental_health_score	socioeconomic_status	school_quality_score	baseline_cognitive_score	random_noise
0	3.299	4.447	140.703	0.497	82.678	118.961	0.682
1	4.974	8.696	142.483	-0.138	81.532	107.864	0.126
0	3.646	7.473	138.472	0.648	74.587	107.073	-0.222
0	6.858	9.413	163.341	1.523	89.436	108.761	2.047
0	7.256	11.372	153.809	-0.234	70.524	104.641	-0.675
0	8.030	13.165	162.330	-0.234	71.178	101.593	-0.403
1	8.973	12.010	182.788	1.579	85.264	124.744	1.992
1	7.896	10.558	171.341	0.767	77.684	114.026	-0.832
1	3.327	6.779	139.325	-0.469	71.438	105.801	-0.550
1	6.106	9.725	156.019	0.543	76.017	100.073	-0.147
0	3.389	6.297	135.484	-0.463	72.374	108.540	0.841
0	3.586	6.279	128.820	-0.466	71.073	97.319	0.208
0	4.956	8.337	162.221	0.242	75.980	123.172	-1.293
0	2.195	1.886	89.753	-1.913	56.064	73.975	-0.532
0	0.179	2.410	103.319	-1.725	58.894	100.110	-0.607
0	5.669	9.725	143.092	-0.562	73.523	96.356	-0.077
1	3.938	5.744	109.827	-1.013	68.901	83.358	0.426
1	7.023	10.333	148.110	0.314	73.604	98.304	0.418
1	8.701	9.950	138.848	-0.908	68.133	87.716	-1.776
1	3.801	6.272	119.439	-1.412	66.707	90.119	1.064
1	3.932	7.938	157.370	1.466	80.048	119.885	0.253
0	4.428	6.926	125.519	-0.226	75.367	92.005	1.385

Data Structure - Relationship between Variables

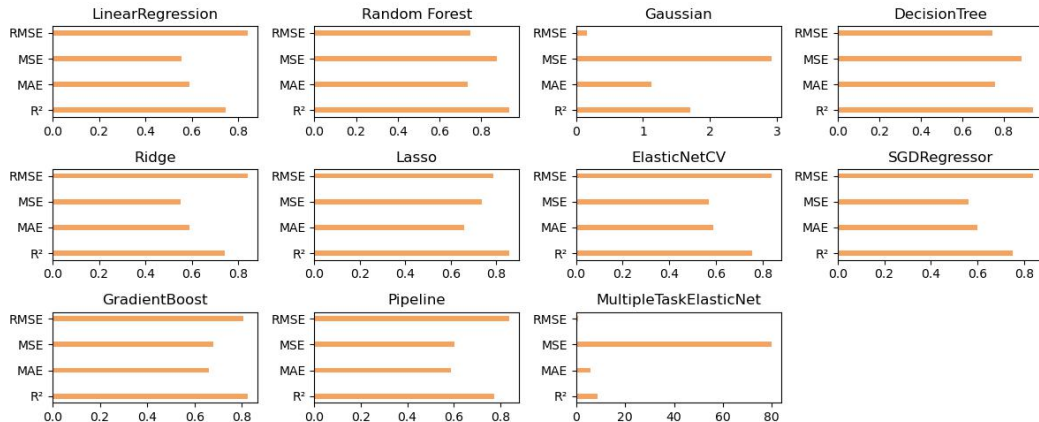
Statistical DAG



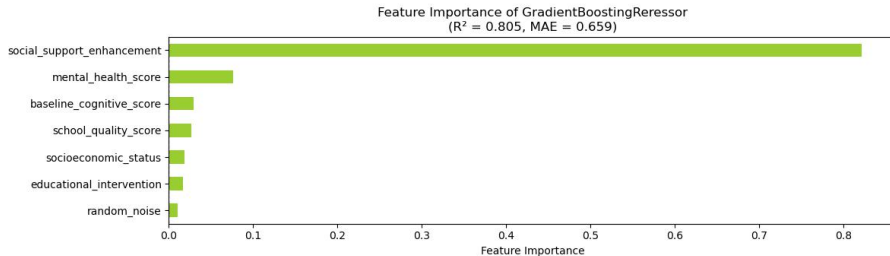
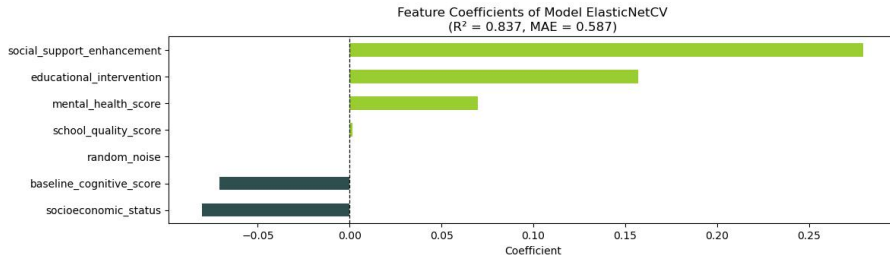
Data Structure - Distribution of Variables



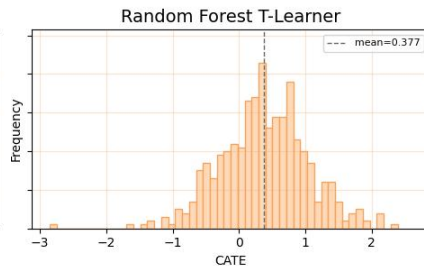
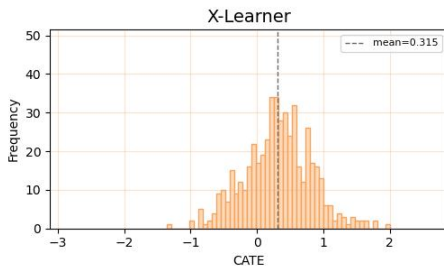
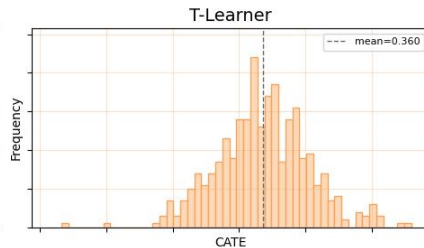
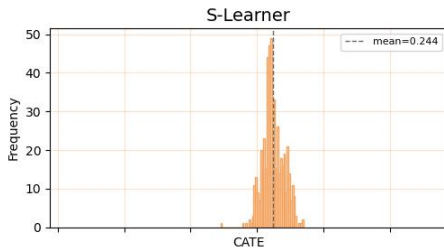
Regression Models - Performances



Regression Models - Feature Importance and Coefficients



T,S,X-Learners and Causal Forests



Final Results

- **Selection Bias:** treated group has higher starting values according to SMD (Social support enhancement - SMD: 0.748, mental health score - SMD: 0.575)
- **Predictive Influence:** social support enhancement dominates both ElasticNetCV (beta = 0.28) and GradientBoosting (importance = 0.82)
- **Heterogeneous Treatment Effect:** CATE distribution of T- and Random Forest T-Learner between $(-3 \text{ and } +2)$ indicates, that the intervention does not affect all individuals equally
- **Validation:** Random noise ≈ 0

